12_Logarithmic Properties & Exponential Rules

1. Exponential Rules (Laws of Exponents)

Exponents show **repeated multiplication**.

Rule 1: Product of Powers

$$a^m \cdot a^n = a^{m+n}$$

Example:

$$2^3 \cdot 2^4 = 2^{3+4} = 2^7 = 128$$

Rule 2: Quotient of Powers

$$\frac{a^m}{a^n} = a^{m-n}(a \neq 0)$$

Example:

$$\frac{5^6}{5^2} = 5^{6-2} = 5^4 = 625$$

Rule 3: Power of a Power

$$(a^m)^n = a^{mn}$$

Example:

$$(3^2)^4 = 3^{2\times4} = 3^8 = 6561$$

Rule 4: Power of a Product

$$(ab)^n = a^n b^n$$

Example:

$$(2 \cdot 5)^3 = 2^3 \cdot 5^3 = 8 \cdot 125 = 1000$$

Rule 5: Negative Exponents

$$a^{-n} = \frac{1}{a^n}$$

Example:

$$4^{-2} = \frac{1}{4^2} = \frac{1}{16}$$

Rule 6: Zero Exponent

$$a^0 = 1 (a \neq 0)$$

Example:

$$7^0 = 1$$

2. Logarithmic Properties

A **logarithm** answers the question:

"To what power must the base be raised?"

$$\log_b a = x \iff b^x = a$$

Property 1: Product Rule

$$\log_b(MN) = \log_b M + \log_b N$$

Example:

$$\log_2(8 \cdot 4) = \log_2 8 + \log_2 4 = 3 + 2 = 5$$

Property 2: Quotient Rule

$$\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N$$

Example:

$$\log_{10}\left(\frac{1000}{10}\right) = \log\ 1000 - \log\ 10 = 3 - 1 = 2$$

Property 3: Power Rule

$$\log_b(M^n) = n\log_b M$$

Example:

$$\log_3(9^2) = 2\log_3 9 = 2 \times 2 = 4$$

Property 4: Log of 1

$$\log_b 1 = 0$$

Example:

$$\log_5 1 = 0$$

Property 5: Log of the Base

$$\log_b b = 1$$

Example:

$$\log_7 7 = 1$$

3. Relationship Between Logs and Exponents

They are **inverse operations**.

$$\log_b(b^x) = x$$
$$b^{\log_b x} = x$$

Example:

$$\log_2(2^6) = 6$$

4. Solved Example (Exam Style)

Solve:

$$\log_2 x + \log_2 8 = 5$$

Step 1: Use product rule

$$\log_2(8x) = 5$$

Step 2: Convert to exponential form

$$2^5 = 8x$$

Step 3: Solve

$$32 = 8x \Rightarrow x = 4$$